Politeness Is Not a Ruler: The Non-Linear Geometry of a Graded Social Trait in LLM Representations

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Abstract

According to the linear representation hypothesis, LLMs encode high-level concepts as directions in activation space. The supporting evidence, however, comes almost entirely from *binary* contrasts, which can show a separating direction exists without testing whether intensity along it is linear. We ask this stronger question for *politeness* (a graded, high-level pragmatic trait with no external scalar) by adding an explicit NEUTRAL level between the poles, the minimal structure that makes linearity falsifiable. We build a dataset of same-content paraphrases varying only in politeness and probe Gemma 2 2B under several complementary geometries. Politeness admits a clean binary axis yet is *not* a single ordered ruler: the levels are linearly orderable, but the adjacent transitions do not align and the neutral level lies off the line joining the poles—linear encoding of a graded trait need not be one-dimensional.

1 Introduction

Linear representations in LLMs. A growing body of mechanistic work suggests that high-level concepts in large language models (LLMs) are encoded as approximately *linear* directions in the residual stream [Elhage et al., 2021, Park et al., 2024]. The idea predates LLMs (word-vector analogies [Mikolov et al., 2013] placed relations such as $\text{KING} - \text{MAN} + \text{WOMAN} \approx \text{QUEEN}$ on linear subspaces) and has since been sharpened for transformers, with linear codes reported for sentiment [Tigges et al., 2023], truthfulness [Marks and Tegmark, 2024], and continuous quantities such as space and time [Gurnee and Tegmark, 2024]. Representation-engineering and steering methods exploit the same structure to amplify or suppress behaviours without fine-tuning [Zou et al., 2023, Turner et al., 2023, Subramani et al., 2022, Rimsky et al., 2024].

Research Goal. Two clusters are collinear in any geometry, so a binary separation shows only that a direction *exists*; it cannot test whether intensity along it is linear—whether the representation behaves like an ordered *ruler*. We ask this stronger question for *politeness*: a high-level pragmatic attribute, graded by construction (a request can be rude, neutral, or deferential), socially constructed rather than anchored to an external scalar, and—unlike sentiment—only weakly carried by individual

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valenced words. Concretely: *is politeness intensity encoded as a single, linearly graded direction, with the neutral level lying proportionally between the two poles?*

Contributions. Two commitments separate our study from prior work. *First, we test graded linearity, not binary separability* adding an explicit NEUTRAL level that makes linearity falsifiable. *Second, we probe a high-level pragmatic trait, not a lexical or externally grounded one.* Both demand strict *content control* so we build the dataset ourselves (Section 3.1) with an independent LLM judge.

2 Background and Related Work

Measuring linearity, not just finding directions. The works above show that concept directions *exist*; our question is about their *geometry*, which raises issues those studies do not face: a linear probe recovering an attribute is weak evidence that the model actually *uses* that direction [Belinkov, 2022, Alain and Bengio, 2016], so difference-of-means estimators and causal steering are preferred for isolating behaviourally relevant axes [Marks and Tegmark, 2024, Rimsky et al., 2024]. Measuring whether intensity is encoded *linearly* is subtler still: cosine similarity in the raw residual stream is not coordinate-free [Park et al., 2024] and the space is strongly anisotropic [Ethayarajh, 2019], so we evaluate directions under several complementary geometries rather than one (Section 3).

Politeness in NLP: behavioural, not representational. Politeness has been studied extensively in NLP, but almost exclusively at the *behavioural* level—phrase-level classifiers [Danescu-Niculescu-Mizil et al., 2013], controllable rewriters [Niu and Bansal, 2018, Madaan et al., 2020], and few-shot style transfer [Reif et al., 2022]—all operating on mode outputs, not internal representations.

Graded representations, and the open question. Tigges et al. [2023] and Konen et al. [2024] recover single directions for sentiment and emotion, but both treat them as bipolar and neither tests whether an intermediate level sits proportionally between the poles. Heinzerling and Inui [2024] come closest—distinguishing *linear* from *monotonic* encodings—but only for externally grounded numeric properties that vary across *entities*, whereas politeness has no external scalar and must be elicited by rewriting the *same* content. Whether a graded *pragmatic* trait forms a one-dimensional ordered ladder thus remains open.

3 Method

3.1 Dataset construction

Our stimulus set must be *content controlled*: items at different politeness levels must differ in intensity and, as far as possible, nothing else. No public corpus offers this at the granularity we need, so we generate the benchmark ourselves with a two-model pipeline: a *generator* (Gemini 2.0 Flash) and an independent *judge* (DeepSeek v4 Flash). Appendix A gives full detail.

Scenarios and ladders. The unit of generation is a *scenario*: a fixed communicative situation that pins down everything except how politely it is expressed—an *intent* (one of ten illocutionary types: request, refusal, disagreement, criticism, bad-news delivery, apology, complaint, reminder, sensitive inquiry, correction), a domain, the speaker–addressee relation, and an *intent target* (the concrete content every rung must convey, e.g. "send the budget spreadsheet by end of day"). From each scenario the generator writes a *ladder*: three sentences, one per signed level (*negative*, *neutral*, *positive*, neutral a genuine zero), conveying the same target and differing only in politeness, with up to three paraphrases per level. Ten intents keep any single act from dominating the geometry; the dataset is English-only, with 72 complete scenarios (633 paraphrases) after filtering.

Confound control and filtering. Because politeness is read from the *difference* between a scenario's rungs, content varying across rungs would be mistaken for trait signal. We guard against this twice over. By construction, paraphrases within a level use distinct *cue families* (courtesy markers, indirectness, gratitude, imposition acknowledgment, impersonal phrasing), and every rung stays within a fixed word-count tolerance, so neither a single marker nor length can cue the level. The judge then applies a *content* check (faithfulness to the target), a *length* check, and—decisively—a *blind*

Estimator	Balanced acc. (raw / centered)	Cosine to mean-diff (centered)
Difference-of-means	0.914 / 0.971	1.00
k-means	0.759 / 0.967	0.996
Logistic regression	0.962 / 0.991	0.787
PCA	0.755 / 0.950	0.966
Random direction	—	0.025

Table 1: Binary POSITIVE-VS-NEGATIVE contrast (last token, layer 13): held-out balanced accuracy under scenario-grouped CV, and the cosine of each estimated axis to the difference-of-means direction after content control. All four estimators separate the poles and align once centered; the GPT-2 sentiment references are 0.80–0.89.

intensity check that re-rates politeness with level labels hidden, keeping a bundle only if its blind per-level means are ordered NEGATIVE $<$ NEUTRAL $<$ POSITIVE with a minimum gap. The ordering is thus recovered by a label-blind rater, not asserted. A bag-of-words baseline already classifies the level at 85.0% (chance 33%)—a few lexical cues carry much of the signal—so we do not claim separability as a result and study the *graded geometry* instead, which it cannot express.

4 Experiments

We probe **Gemma 2 2B** [Gemma Team, 2024], reading the residual stream at layer 13 of 26—mid-network, where high-level features tend to concentrate [Tigges et al., 2023]—at the *last token* of each paraphrase, the convention in steering work [Zou et al., 2023, Tigges et al., 2023]; mean-pooling reproduces every result (Section C). One methodological choice matters throughout: residual activations are dominated by each scenario's content, so we treat `scenario_id` as a block, subtracting its across-level mean and evaluating held-out under `GroupKFold`. This *blocked* estimator is the one we interpret; a naive pooled baseline (Section B) shows how much content otherwise inflates the numbers. We re-measure every alignment under several whitenings and against a within-scenario permutation null, and report nothing the noise floor can explain (Section B). Because a bag-of-words classifier already labels the level at 85.0% (chance 33%), the question is never whether politeness is *decodable*, but whether it is laid out as a single linear scale. We ask this in three steps of increasing strength: does a binary axis exist (Section 4.1); do the three ordered levels fall on one straight line (Section 4.2); and does a single direction steer politeness across every context (Section 4.3)? All analyses use the 72 complete scenarios (633 paraphrases).

4.1 A clean binary axis exists

Restricting the ladder to its poles reproduces, for politeness, the linear separability Tigges et al. [2023] reported for sentiment. The two poles project onto the difference-of-means axis with ROC-AUC 0.98 and Cohen's $d = 3.0$, and a logistic probe classifies held-out scenarios at 96.2% balanced accuracy (97.1% after within-scenario centering)—above the GPT-2 sentiment reference (0.80–0.89) and far above a random-direction floor (cosine ≈ 0.02). Four independent estimators (difference-of-means, k-means, logistic, PCA) recover essentially the same axis once content is controlled (Table 1). Both adjacent half-steps are separable on their own too (NEG–NEU 88%/95% raw/centered; NEU–POS 76%/85%), so each transition is individually linearly recoverable. Binary separability is therefore not in question—which is exactly why it cannot, by itself, establish a graded *ruler*.

4.2 The graded ladder is ordered but bent

Adding NEUTRAL lets us ask whether the three levels lie, *in order*, along one line. They are reliably *ordered*: against the endpoint (negative$\rightarrow$positive) axis the blocked Spearman $\rho = 0.84$ and Kendall $\tau = 0.70$ (near their structural ceilings of 0.94/0.83), a cross-validated ridge decodes rank at $R^2 = 0.85$, and 81% of scenarios are monotone—every metric beats its within-scenario permutation null at $p \approx 0.01$. But the levels are *not collinear*. The two transition directions (NEU$-$NEG and POS$-$NEU) point apart rather than together: their cosine is -0.23 (a straight ladder predicts $+1$), the apex angle at neutral is 77° (straight $= 180^\circ$), and the neutral centroid sits a normalised 0.61 of the pole-to-pole span off the midpoint (straight $= 0$). The bend survives a stringent null: forcing neutral onto the true midpoint and re-injecting the empirical paraphrase noise yields a null step-cosine

Descriptor	Observed (95% CI)	Straight ladder	Noise-null p
Step cosine	-0.23 $[-0.33, -0.13]$	$+1$	0.001
Apex angle	$77°$ $[71, 83]$	$180°$	—
Midpoint residual	0.61 $[0.56, 0.68]$	0	0.001

Table 2: Triple geometry at layer 13 (72 scenarios, scenario-bootstrap CIs). All three descriptors are far from the straight-ladder reference, and the bend cannot be reproduced by noise on a true straight ladder.

of $+0.92{\pm}0.01$ and midpoint residual $0.10{\pm}0.01$, so measurement noise cannot manufacture the observed values ($p = 0.001$, $\approx$tens of σ; Table 2). The negative cosine persists under all four inner products (-0.14 to -0.23) and after disattenuation. The ladder is thus a *V-shape*: orderable, yet not laid out along a single straight ruler.

4.3 No universal steering direction

Steering presumes a *single* direction that works everywhere. Averaged over scenarios, a global politeness axis is indeed highly reproducible (split-half Spearman–Brown $R = 0.94$). But reproducible-on-average is not the same as universal: that one vector recovers only about a fifth of any individual scenario's neg→pos contrast (mean capture $\cos^2 = 0.20$, IQR 0.13–0.26). The shortfall is not noise. The off-line bend has its own reliable, shared direction—a "markedness" axis on which neutral lands on the same side in 96% of scenarios ($R = 0.89$)—yet the shared (valence, markedness) plane still captures only 0.21 of within-scenario variance, which is already 94% of all the variance any plane could capture (Section B). The rest is genuine, scenario-specific structure: contrasts align more within a communicative intent (0.25) than across intents (0.16), and the pattern is stable with depth—capture plateaus near 0.20 from layer $\sim$12 and never approaches 1. A reproducible *mean* axis thus coexists with contrasts that are mostly scenario-specific: an average over dispersion, not a single shared mechanism.

What this means for "linear representation." The three results separate senses that binary studies conflate. *Ordinal* linearity **holds**: the levels sit in order along a recoverable axis. *Metric* linearity **fails**: neutral is not at the midpoint and the steps do not align, so the scale is not one straight ruler. *Universal-direction* linearity **fails**: no single vector reproduces the contrast across contexts. Linear *decodability* of a graded pragmatic trait therefore does not imply a one-dimensional, steerable encoding—a distinction with direct consequences for activation steering, which a single politeness vector moves only $\sim$20% of the way in any given context.

5 Conclusion

We asked whether a graded pragmatic trait—politeness—is encoded as a single linear ruler in Gemma 2 2B. Using a content-controlled, blind-validated ladder with an explicit neutral level, we find a clean binary axis but a ladder that is *ordered without being collinear* (neutral lies off the pole-to-pole line; adjacent steps anti-align at a $77°$ apex) and that admits *no universal steering direction* (a reproducible mean axis captures only $\sim$20% of any scenario's contrast). Linear decodability does not entail one-dimensional, steerable geometry. Our evidence is one model, one trait, and three discrete levels with $\sim$3 paraphrases per cell—which limits how cleanly reliable scenario-specific structure can be separated from estimation noise; finer level grids, more paraphrases per cell, and causal intervention tests are the natural next steps.

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A Dataset construction details

This appendix gives the full specification of the corpus summarised in Section 3.1. The pipeline runs in four stages—scenario generation, paraphrase generation, judging/filtering, and activation extraction—driven by two models, a *generator* (Gemini 2.0 Flash) and a *judge* (DeepSeek v4 Flash).

A.1 Construct rubric

Politeness is defined on a signed three-point scale as the mitigation of face threat, deference, and social consideration, with neutral a true zero (no courtesy and no rudeness markers) and the poles equal-and-opposite departures from it. **Negative** is dismissive, curt, or impatient, with rudeness from tone and framing rather than profanity, threats, or personal attacks (which would be lexical giveaways); **neutral** is plain and even-toned; **positive** is respectful and mitigated—appreciation, deference, acknowledgment of the imposition—through framing rather than added words. Held constant across the three levels of a scenario: the intent target, named entities, deadlines, the polarity of the act, the amount imposed, and the length.

A.2 Scenario schema and intents

Each scenario is a structured record specifying the *intent* (one of ten: *request, refusal, disagreement, criticism_or_feedback, bad_news_delivery, apology, complaint, reminder, inquiry_sensitive, correction*), a *domain* (e.g. workplace, restaurant, healthcare), the *audience relation* between speaker and addressee (e.g. peer, subordinate, service agent), a *communicative goal*, the *intent target* (the concrete content every rung must convey), and a *target word count*. Each intent carries its own invariance constraint—for example, a refusal must remain a refusal and may not become a partial acceptance. Intents are drawn from a shuffled queue, so the generated set is balanced across the ten.

A.3 Generation and judging

Models. Two models drive the pipeline (Table 3): the generator produces both scenarios and paraphrases, and the judge performs all filtering, including the blind intensity rating. A configuration check refuses to run if generator and judge share a model family, so generation and validation are never performed by the same model.

Role	Model	Temperature
Generator (scenarios + paraphrases)	`google/gemini-2.0-flash-001`	0.5
Judge (filtering + blind scoring)	`deepseek/deepseek-v4-flash`	0.0
Probed model (activation extraction)	`google/gemma-2-2b`	—

Table 3: Models used in dataset construction. Generator and judge are required to be from different families; the probed model is independent of both.

Acceptance criteria. After generation, each scenario's bundle of paraphrases passes through three judge checks, with parameters fixed for the run:

- **Context (content preservation).** The judge scores each paraphrase in $[0, 1]$ for how faithfully it conveys the scenario's intent target—explicitly ignoring politeness and fluency—and paraphrases must meet a minimum acceptance score of 0.70.

- **Trait intensity (blind monotonicity).** The judge re-rates each paraphrase's politeness in $[0, 1]$ with level labels *hidden*: texts appear under opaque codes (`q000`, `q001`, …) in shuffled order, with no indication of which belong together. A bundle is kept only if the blind per-level means are ordered NEGATIVE $<$ NEUTRAL $<$ POSITIVE with a minimum adjacent gap of 0.10.
- **Length.** Per-level word counts must not diverge by more than a ratio of 1.15, so length cannot cue the level.

A near-duplicate guard additionally removes paraphrases whose content overlaps an earlier one above a Jaccard threshold. The intent queue and the blind-scoring order use a fixed random seed for reproducibility.

A.4 Resulting corpus

From 100 generated scenarios (900 paraphrases), 633 passed all three checks (a 70.3% acceptance rate); the 267 rejections were dominated by the trait-intensity gate (163), followed by length (72) and context (32). The filtered corpus spans 72 scenarios for which all three levels survive, balanced across the ten intents, with per-level counts of 212 negative, 214 neutral, and 207 positive paraphrases (Table 4). Sentence length is well matched across levels (mean word counts 16.9, 17.4, 17.7), confirming the length control held in the final data. Cue families are recorded as free-text labels and span a broad vocabulary rather than a fixed set, reflecting the diversification target.

Quantity	Value
Scenarios generated	100
Paraphrases generated	900
Paraphrases passing all checks	633 (70.3%)
rejected: trait intensity / length / context	163 / 72 / 32
Complete scenarios (all three levels) in filtered set	72
Paraphrases by level (neg / neu / pos)	212 / 214 / 207
Mean length by level (neg / neu / pos, words)	16.9 / 17.4 / 17.7
Bag-of-words classifier accuracy (5-fold CV)	85.0% (chance 33%)

Table 4: Dataset statistics for politeness.

Lexical baseline. The bag-of-words baseline uses unigram and bigram counts ($\min$ document frequency 2) with logistic regression under stratified 5-fold cross-validation, predicting the level from raw text. It reaches 85.0% accuracy (chance 33%, $n = 633$): politeness is strongly lexically cued, so we report this not as a rejection criterion but as a floor the geometric analysis must exceed to claim structure beyond surface keywords. The most discriminative n-grams per level are intuitive—NEGATIVE: *immediately, no, late, wrong, seriously, not*; NEUTRAL: *please, scheduled, soon, should*; POSITIVE: *appreciate, hope, thanks, perhaps, might*—confirming that the surface signal is exactly the courtesy and curtness vocabulary one would expect.

A.5 Activation extraction

Activations are taken from the probed model's *prefill* pass over each paraphrase—no text is generated—at every layer in 1–22, and cached. Each paraphrase is reduced to one vector per layer by token pooling: the *last* content token (our primary setting) or the *average* over content tokens (excluding the beginning-of-sequence, end-of-sequence, and padding tokens), with both poolings stored. Paraphrases sharing a (trait, level, scenario) key are averaged. The pooling-invariant unembedding covariance, used for the causal-geometry analysis, is computed once over the model's full output-embedding matrix and stored alongside the activations.

A.6 Example bundle

B Estimation, geometry, and significance

Blocked vs. naive-pooled estimators. Residual activations carry large variance unrelated to politeness, dominated by the invariant content of each scenario. Since `scenario_id` fixes that content, we

Field	Value
`scenario_id`	`politeness-request-001`
Intent / domain	request / workplace
Intent target	send the latest budget spreadsheet by end of day
NEGATIVE	"Is it too much to ask for the budget spreadsheet by day's end? Get it done."
NEUTRAL	"Please provide the latest budget spreadsheet before the end of the day; it's needed today."
POSITIVE	"Would you mind sending over the latest budget spreadsheet by the end of the day, please?"

Table 5: One scenario (`politeness-request-001`) realised at all three levels. All three rungs convey the same intent target—*send the latest budget spreadsheet by end of day*—and differ only in politeness.

treat the scenario as a block and report every metric two ways. The *blocked* (fixed-effects) estimator subtracts each scenario's across-level mean before any geometry is measured and evaluates held-out under `GroupKFold`, so a scenario's levels never split across train and test; it is the estimator we interpret. The *naive pooled* estimator pools points by level only and serves as a baseline that exposes how much the content confound inflates the numbers—blocking raises the Spearman ρ from 0.77 to 0.84 and the probe R^2 from 0.77 to 0.85, and the midpoint residual moves from a pooled 0.61 to a 0.72 per-scenario median.

Four inner products and disattenuation. A near-zero or near-one raw cosine can be an artefact of coordinates rather than a fact about politeness, so we re-measure the transition-direction alignment under four inner products: raw Euclidean; diagonal (anisotropy) whitening; the causal whitening of Park et al. [2024], from the inverse square root of the unembedding covariance over the full vocabulary; and Mahalanobis whitening from the Ledoit–Wolf-shrunk pooled within-level covariance. The adjacent-step cosine stays negative in all four (-0.23 Euclidean, -0.20 anisotropy, -0.21 causal, -0.14 Mahalanobis), and disattenuating by split-half reliability (300 random halves, rescaled by Spearman–Brown) leaves it negative (-0.25 to -0.14). The bend is thus neither a coordinate nor an estimation-noise artefact.

Within-scenario permutation null. To test whether the geometry could arise by chance we shuffle the three level labels *inside* each scenario, holding the block structure intact, and recompute every metric over N permutations; a global shuffle would also destroy content and topic structure, a much weaker null. We report add-one Monte-Carlo p-values, $p = (1 + \#\{\text{null beyond observed}\})/(1 + N)$, with the appropriate tail per metric (lower for the midpoint residual, upper otherwise). All six ordinal metrics clear the null at $p \approx 0.01$.

Lexical baseline and layer robustness. A bag-of-words classifier (unigram and bigram counts, $\min$ document frequency 2, logistic regression, stratified 5-fold CV) predicts the level at 85.0% (chance 33%): politeness is strongly lexically cued, so we treat this as a floor the geometry must exceed, not a result. The most discriminative n-grams are the expected courtesy/curtness vocabulary (NEG: *now, immediately, no, late*; NEU: *please, scheduled, should*; POS: *appreciate, perhaps, hope, thanks*). Repeating the direction-agreement, alignment, and full-distribution analyses across layers 1–22 gives the same qualitative picture throughout: the ordinal structure and the off-line bend are properties of the model, not of the fixed layer.

Steering and the shared-plane ceiling. *Steering capture* is the squared cosine between a scenario's neg$\to$pos contrast and the global difference-of-means axis—the fraction of that contrast a single vector reproduces; we report its mean and IQR over scenarios and across layers. The *shared plane* is spanned by the global valence axis and the markedness (off-line bend) axis; its captured fraction (0.21) should be read against the ceiling set by each scenario's own best two-dimensional fit, the top-two principal components of the within-scenario centred triples (0.22). That the shared plane reaches 94% of this ceiling, while a high participation ratio (26.5) shows the per-scenario planes are spread over many dimensions, together mean the residual is real scenario-specific orientation rather than variance a better shared plane could absorb.

C Mean-pooling robustness

The main text reads the last-token residual. As a robustness check we re-run *every* analysis—binary, graded, and steering—under *mean-pooling*, averaging the layer-13 residual over each paraphrase's content tokens (excluding the BOS, EOS, and padding positions). The mean-pooled bundle is precomputed under `data/<timestamp>/representations/avg_token/` and consumed by the same scripts (a `TOKEN_POOLING` constant selects the branch); figures land under `results/<timestamp>/.../avg_token/`.

Every conclusion holds, and is if anything sharper than at the last token. The binary poles separate at least as well (logistic balanced accuracy 0.976 raw, ≥ 0.99 after centering). The ladder is ordered (blocked Spearman $\rho = 0.90$, Kendall $\tau = 0.76$, probe $R^2 = 0.92$) yet bent: step cosine -0.29, apex $73°$, midpoint residual 0.66, with the same noise null rejected at $p = 0.001$. And there is still no universal steering direction: the mean axis is reproducible ($R = 0.96$) but captures only $\cos^2 = 0.25$ of any scenario's contrast, the shared plane captures 0.26 (94% of the two-dimensional ceiling), and the tilt is intent-structured (within 0.39 vs. between 0.22). Reading a single token is therefore not what produces the non-linearity.

D Supplementary figures and tables

All figures below are produced by the analysis scripts and stored under `results/20260530_001930/` (last-token pooling). Table 6 breaks the triple geometry down by communicative intent: every one of the ten intents is bent (apex 66–89°, none near $180°$), with requests and disagreements the straightest and complaints, corrections, and refusals the most bent.

Intent	n	Apex°	Step cosine	Midpoint resid.
apology	8	78.2	−0.20	0.60
bad-news-delivery	4	72.3	−0.30	0.64
complaint	7	66.2	−0.40	0.76
correction	9	67.9	−0.38	0.73
criticism-or-feedback	8	67.9	−0.38	0.73
disagreement	6	82.9	−0.12	0.56
inquiry-sensitive	4	79.8	−0.18	0.60
refusal	8	73.7	−0.28	0.65
reminder	6	73.2	−0.29	0.66
request	12	88.6	−0.02	0.51
ALL	72	76.8	−0.23	0.61

Table 6: Triple geometry pooled within each communicative intent (last token, layer 13). Apex angles stay well below the straight-ladder $180°$ for every intent.

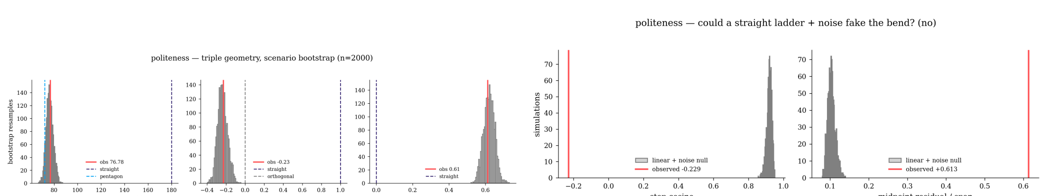

Figure 1: Triple geometry. *Left:* bootstrap distributions of the geometry descriptors against the straight-ladder and pentagon references. *Right:* the linear-ladder noise null—noise on a straight ladder cannot reproduce the observed step cosine or midpoint residual.

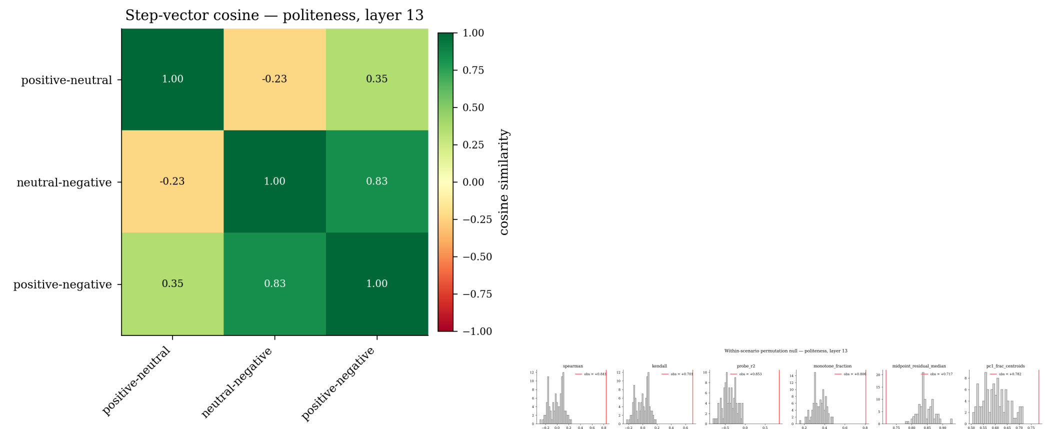

Figure 2: *Left:* cosines among the three step vectors; the two adjacent half-steps anti-align (-0.23). *Right:* the within-scenario permutation null for the six ordinal metrics.

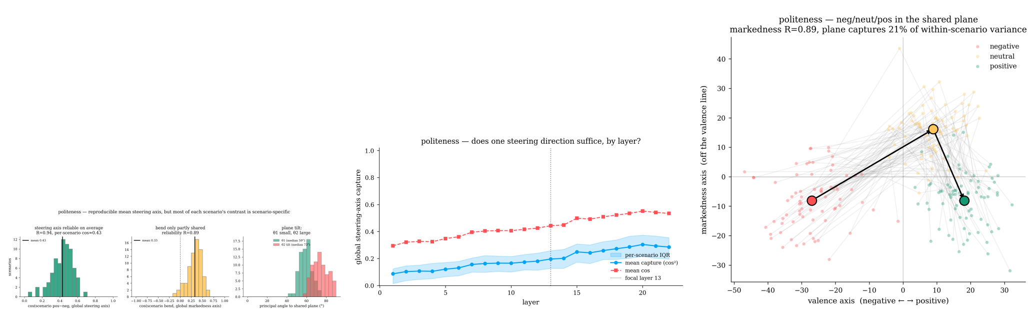

Figure 3: No universal steering direction. *Left:* per-scenario alignment to the global axis and plane-tilt angles. *Middle:* steering capture ($\cos^2$) across layers, plateauing near 0.20. *Right:* the shared (valence, markedness) plane.